THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE TOWN COUNCIL

FORM FOUR PRE- MOCK EXAMINATION APRIL -2024

032/2 **CHEMISTRY 2A**
(ACTUAL PRACTICAL)

TIME: 2:30 Hours **23rd April 2024**

INSTRUCTIONS

1. This paper consists two questions
2. Each question carries 25 marks
3. Attempt all question carefully
4. Where necessary you may use the following constant
5. Write your number on every page of your answer sheet

Atomic masses: H=1, O=16, N=14, Na=23, K=39, Ca=40.

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1. **You are provided with the following solutions**

- BA1 which is 0.2 M Hydrochloric acid solution
- BA2 which is a salt solution made by dissolving 10.6g of M_2CO_3 in litre of solution
- Methyl orange

Procedures,

Pipette 25ml of BA2 into clean conical flask. Add two or three drops of an indicator from the provided. Titrate the solution BA1 from the burette. Repeat the titration to obtain consistent titres. Record your results in a tabula form.

Questions

i. Why methyl indicator was used

ii. Find the mean titre volume of BA1 used in this experiment

iii. calculate

   a) the number of moles of solution BA1

iv. calculate

   a) Molarity of solution BA2 solution

   b) Calculate the relative atomic mass of M in M_2CO_3

2. **You are provided with substance X which contains one cation and one anion. Cary out the unguided experiment to identify those cation and anions present in it**

Experiment	Observation	Inferences

Conclusion

a) The cation present is____ and anion present is _____

b) The chemical name of compound X

c) The UPAC name of compound X